

SLIC Index V3

Methodology Technical Paper

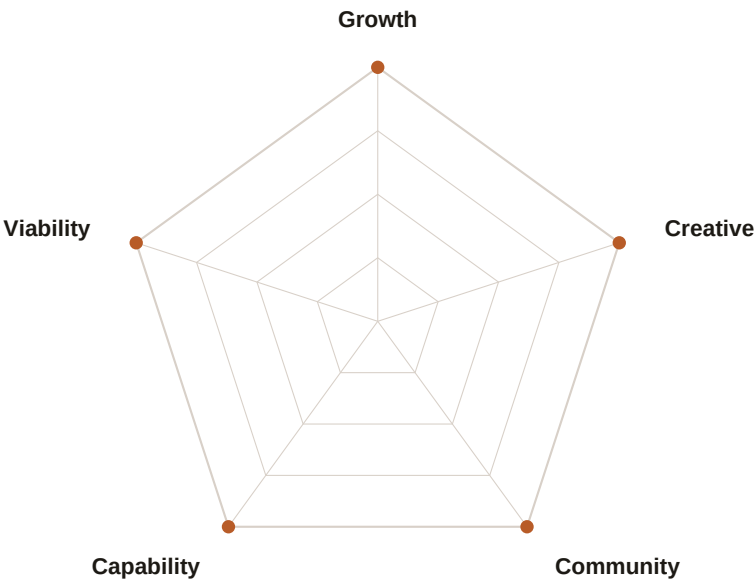
April 2026

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Published by SLIC and ReTL (The Reason to Live Company)

In partnership with DEPA Thailand and PMU-A

This paper documents the complete scoring methodology for the SLIC Index V3: indicator definitions, normalization anchor tables, aggregation formula, coverage grades, and the full published rankings for 160 cities.



Five pillars. Fifty-point symmetry. Absolute anchors.

Executive Summary

The Smart and Liveable Cities Index (SLIC) V3 is a transparent, open-source city ranking system measuring quality of life for people in the process of building lives — not tourists, not expats on hardship pay, not global capital seeking returns. It covers 160 published cities across an Asia-Pacific-centred but globally inclusive dataset, scored on 35 indicators grouped into five pillars.

Five pillars:

Pillar	Weight	What it measures
Growth	25%	Economic dynamism, disposable income, startup density, civic freedom
Viability	22%	Safety, air quality, healthcare, housing affordability, water access
Capability	18%	Transit, digital infrastructure, education, renewable energy, walkability
Community	15%	Tolerance, LGBTQ+ rights, gender equality, social trust, income inequality
Creative	20%	Cultural venues, culinary diversity, arts funding, creative employment

Absolute scoring philosophy:

- All indicators are normalized against fixed absolute benchmarks — adding city #501 never changes any existing score.
- Higher normalized scores always mean better urban outcomes; harmful indicators are reverse-scored inside the normalization step.
- The Adjusted Mazziotta–Pareto Index (AMPI) aggregation formula penalizes pillar imbalance: a city with one excellent pillar and four poor ones scores lower than a city with five moderate pillars.
- Missing data is never imputed. Cities with fewer than the minimum required indicators receive a coverage penalty (–5 or –15 points) and a visible grade flag.
- Every source is named. Every score is traceable back to a raw data point, a normalization function, and a published source.

1.

Scoring Framework

1.1 Piecewise Linear Normalization

Each raw metric value is mapped to a 0–100 score using a fixed set of absolute anchor points. Between anchor points, scores are linearly interpolated. Values beyond the outer anchors are clamped to 0 or 100. For lower-is-better indicators the anchor mapping is reversed so that lower raw values produce higher normalized scores.

$$s_m(c) = 100 \times (\text{winsor}(x_m(c), p_{05_m}, p_{95_m}) - p_{05_m}) / (p_{95_m} - p_{05_m})$$

Where:

`x_m(c)` = raw metric value for city `c` on metric `m`
`p05_m` = 5th percentile across the reference set, frozen at first publication
`p95_m` = 95th percentile, likewise frozen
`winsor()` = clamps to `[p05, p95]` before normalising
`s_m(c)` = normalized metric score, clamped to `[0, 100]`

For lower-is-better indicators (homicide, housing burden, hours worked, etc.), the numerator is inverted so that lower raw values map to higher normalized scores. Anchors are frozen – adding city #501 never retroactively changes cities #1–500.

1.2 Pillar Aggregation (AMPI)

Within each pillar, constituent metric scores are aggregated using the Adjusted Mazziotta–Pareto Index (AMPI). The AMPI penalizes uneven performance: a city with high variance across its metrics scores lower than one with similar mean but lower variance. This makes it impossible to compensate for a catastrophic metric score by excelling elsewhere.

$$\text{pillar_score} = \Sigma(s_m \times \alpha_{(p,m)}) / \Sigma \alpha_{(p,m)} \quad [\text{weighted mean}]$$

Where:

`α(p,m)` = published metric weight within pillar `p` (see Chapter 7)
`s_m` = normalized metric scores for that city, non-null only

Pillar weights are designed so that $\Sigma \alpha_{(p,m)}$ within each pillar equals the pillar's public weight: Pressure 25, Viability 22, Capability 18, Community 15, Creative 20 (sum = 100).

1.3 Overall SLIC Score

Overall SLIC score is aggregated across the five pillar scores using the Adjusted Mazziotta–Pareto Index (AMPI):

`μp` = weighted mean of pillar scores (weights `wp` = 25/22/18/15/20)
`σ2p` = $\Sigma(w_p \times (\text{pillar}_p - \mu_p)^2) / \Sigma w_p$ [weighted variance]
`AMPI` = `μp` – `σ2p` / `μp`

$$\text{SLIC}(c) = \max(0, \text{AMPI}(c) - \text{coverage_penalty}(c))$$

AMPI penalises pillar imbalance. A city with four strong pillars and one catastrophic pillar scores lower than a city with five moderate

pillars – even if their weighted means are identical. This is the single distinctive claim SLIC makes versus other indices: you cannot compensate for a broken pillar with excellence in another. AMPI is applied at the pillar combination level (n=5, stable variance) rather than within pillars (where missing metrics would make variance unreliable).

1.4 PPP-Adjusted Disposable Income (DI_PPP)

The signature SLIC metric. Measures residual monthly income after all essential costs, converted to PPP-adjusted USD so cities in different economic contexts are comparable.

$$\text{DI_PPP}(c) = \left[\begin{aligned} &\text{GrossIncome}(c) \times (1 - \text{TaxRate}(\text{country}(c))) \\ &- \text{Rent}(c) \\ &- \text{Utilities}(c) \\ &- \text{Transit}(c) \\ &- \text{Internet}(c) \\ &- \text{Food}(c) \end{aligned} \right] \div \text{PPP_PrivateConsumptionFactor}(\text{country}(c))$$

2.

Pillar 1: Growth

Economic dynamism and opportunity (25% weight)

The Growth pillar measures whether a city creates the conditions for people to build economically productive lives: disposable income after costs, startup and innovation density, civic freedom, and broad economic momentum.

G1 — Real GDP Growth (5-year avg)

Unit: % per annum · Direction: ↑ Higher is better · Source: IMF World Economic Outlook; OECD Regional Statistics

Raw value	Score
≤ 0	0
1.0	25
2.5	50
4.0	75
≥ 6.0	100

G2 — Startup Density

Unit: per 100,000 population · Direction: ↑ Higher is better · Source: Crunchbase; Dealroom

Raw value	Score
≤ 5	0
20	25
50	50
100	75
≥ 200	100

G3 — VC Investment Intensity

Unit: USD per capita, 3-year avg · Direction: ↑ Higher is better · Source: Crunchbase; PitchBook

Raw value	Score
≤ 5	0
50	25
200	50
500	75
≥ 1,500	100

G4 — Ease of Doing Business

Unit: 0–100 index · Direction: ↑ Higher is better · Source: World Bank B-READY Index

Direct passthrough; no transformation required.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

G5 — Civic Freedom

Unit: 0–100 composite · Direction: ↑ Higher is better · Source: Freedom House (0.6) + V-Dem Liberal Democracy Index (0.4)

Composite: $0.6 \times \text{FH_rescaled}(0\text{--}100) + 0.4 \times \text{V-Dem_rescaled}(0\text{--}100)$. Direct passthrough after composite.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

G6 — Patent Applications

Unit: per 100,000 population, 3-year avg · Direction: ↑ Higher is better · Source: World Intellectual Property Organization (WIPO)

Raw value	Score
≤ 2	0
15	25
40	50
80	75
≥ 150	100

G7 — High-Skill Employment

Unit: % ISCO categories 1–3 · Direction: ↑ Higher is better · Source: International Labour Organization (ILO); LinkedIn Workforce Insights

Raw value	Score
≤ 10	0
20	25
30	50
40	75
≥ 55	100

3.

Pillar 2: Viability

Lived sustainability and safety (22% weight)

The Viability pillar measures whether a city is liveable at the ground level: personal safety, air quality, healthcare, housing cost burden, water access, climate stability, and demographic vitality.

V1 — Homicide Rate

Unit: per 100,000 population · Direction: ↓ Lower is better · Source: UNODC International Homicide Statistics

Lower is better; anchor order is reversed.

Raw value	Score
≥ 50	0
20	25
5	50
1.0	75
≤ 0.3	100

V2 — Air Quality (PM2.5)

Unit: µg/m³ annual mean · Direction: ↓ Lower is better · Source: WHO Global Air Quality Database; IQAir World Air Quality Report

Lower is better; reversed anchors.

Raw value	Score
≥ 80	0
50	25
25	50
10	75
≤ 5	100

V3 — Healthcare Access & Quality

Unit: HAQ Index 0–100 · Direction: ↑ Higher is better · Source: IHME / Lancet Healthcare Access and Quality Index

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

V4 — PPP-Adjusted Disposable Income

Unit: USD/month residual · Direction: ↑ Higher is better · Source: Derived: Numbeo cost data + World Bank PPP factors

Non-linear anchors reflect diminishing returns above USD 2,000/month.

Raw value	Score
≤ 0	0
200	15
500	35
1,000	55
2,000	75
≥ 4,000	100

V5 — Housing Burden

Unit: % of gross income · Direction: ↓ Lower is better · Source: Numbeo housing cost surveys; OECD Affordable Housing Database

Lower burden = higher score.

Raw value	Score
≥ 70	0
50	25
30	50
20	75
≤ 10	100

V6 — Water & Sanitation

Unit: % population with safely managed access · Direction: ↑ Higher is better · Source: WHO/UNICEF Joint Monitoring Programme

Raw value	Score
≤ 50	0
70	25
85	50
95	75
≥ 99	100

V7 — Peace & Stability

Unit: Global Peace Index (inverted) · Direction: ↑ Higher is better · Source: Institute for Economics and Peace (IEP)

Formula: score = max(0, min(100, (4 – GPI) / 3 × 100)).

Raw value	Score
GPI ≥ 3.5 → 0	0
GPI 3.0 → 17	~17
GPI 2.0 → 50	50
GPI 1.5 → 83	~83
GPI ≤ 1.2 → 100	100

V8 — Birth Rate

Unit: per 1,000 population · Direction: ↑ Higher is better · Source: UN Population Division; national statistics offices

Interpreted as demographic optimism signal, not population policy judgment.

Raw value	Score
≤ 4	0
7	25
10	50
14	75
≥ 20	100

4.

Pillar 3: Capability

Infrastructure and systems quality (18% weight)

The Capability pillar measures urban infrastructure quality: transit coverage, digital connectivity, digital governance, education quality, walkability, cycling infrastructure, and renewable energy share.

C1 — Transit Coverage

Unit: % population within 500m of stop · Direction: ↑ Higher is better · Source: GTFS feeds; ITDP Transit-Oriented Development Database

Raw value	Score
≤ 5	0
20	25
40	50
65	75
≥ 85	100

C2 — Internet Speed

Unit: Mbps median fixed broadband · Direction: ↑ Higher is better · Source: Ookla Speedtest Global Index; M-Lab NDT

Raw value	Score
≤ 5	0
25	25
75	50
150	75
≥ 300	100

C3 — Digital Government

Unit: UN E-Government Development Index × 100 · Direction: ↑ Higher is better · Source: UN Department of Economic and Social Affairs

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

C4 — Education Quality

Unit: PISA average score · Direction: ↑ Higher is better · Source: OECD PISA; UNESCO Institute for Statistics (tertiary fallback)

Tertiary enrollment fallback anchors: ≤5%=0, 15%=25, 25%=50, 40%=75, ≥55%=100.

Raw value	Score
≤ 350	0
400	25
450	50
500	75
≥ 550	100

C5 — Walkability

Unit: Walk Score 0–100 · Direction: ↑ Higher is better · Source: Walk Score; OpenStreetMap pedestrian network analysis

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

C6 — Cycling Infrastructure

Unit: km of protected lanes per 100,000 pop · Direction: ↑ Higher is better · Source: OpenStreetMap CycLOSM layer; Copenhagenize Index

Raw value	Score
≤ 1	0
5	25
15	50
30	75
≥ 50	100

C7 — Renewable Energy Share

Unit: % of electricity from renewables · Direction: ↑ Higher is better · Source: CDP Cities; IRENA; national grid operators

Raw value	Score
≤ 2	0
15	25
35	50
60	75
≥ 90	100

5.

Pillar 4: Community

Belonging, tolerance, and social equity (15% weight)

The Community pillar measures whether a city is genuinely open: LGBTQ+ rights, religious freedom, immigrant integration, income inequality, social trust, gender equality, and weekend retention (a proxy for residents choosing to stay).

O1 — LGBTQ+ Rights

Unit: Composite 0–100 · Direction: ↑ Higher is better · Source: ILGA World; Equaldex

Composite: same-sex marriage (25pts) + anti-discrimination law (25pts) + gender recognition (15pts) + no criminalization (20pts) + social acceptance survey (15pts).

Raw value	Score
0	0
25	25
50	50
75	75
100	100

O2 — Religious Freedom

Unit: Pew GRI inverted · Direction: ↑ Higher is better · Source: Pew Research Center Global Restrictions Index

Formula: score = max(0, min(100, (10 – GRI) / 10 × 100)).

Raw value	Score
$GRI \geq 9 \rightarrow 0$	0
$GRI 7 \rightarrow 22$	~22
$GRI 5 \rightarrow 44$	~44
$GRI 2 \rightarrow 78$	~78
$GRI \leq 0.5 \rightarrow 95$	~95

O3 — Immigrant Integration

Unit: Employment gap (percentage points) · Direction: ↓ Lower is better · Source: OECD International Migration Outlook; national labour force surveys

Lower employment gap = higher score.

Raw value	Score
≥ 30 pp	0
20 pp	25
10 pp	50
5 pp	75
≤ 1 pp	100

O4 — Income Inequality (Gini)

Unit: Gini coefficient 0–1 · Direction: ↓ Lower is better · Source: World Bank PovcalNet; OECD Income Distribution Database

Lower Gini = higher score.

Raw value	Score
≥ 0.60	0
0.45	25
0.35	50
0.28	75
≤ 0.22	100

O5 — Social Trust

Unit: % saying most people can be trusted · Direction: ↑ Higher is better · Source: World Values Survey; Gallup World Poll

Raw value	Score
≤ 5%	0
15%	25
30%	50
50%	75
≥ 70%	100

O6 — Gender Equality

Unit: UNDP GII inverted · Direction: ↑ Higher is better · Source: UNDP Human Development Reports — Gender Inequality Index

Formula: score = $(1 - \text{GII}) \times 100$.

Raw value	Score
GII 1.0 → 0	0
GII 0.75 → 25	25
GII 0.5 → 50	50
GII 0.25 → 75	75
GII 0.0 → 100	100

O7 — Weekend Retention

Unit: Weekend / weekday mobility ratio · Direction: ↑ Higher is better · Source: Mobile analytics platforms; city-level aggregated movement data

Ratio above 1.0 indicates residents actively choose to stay or return on weekends.

Raw value	Score
≤ 0.70	0
0.80	25
0.90	50
1.00	75
≥ 1.10	100

6.

Pillar 5: Creative

Cultural richness and creative economy (20% weight)

The Creative pillar measures lived cultural richness for residents, not tourists: cultural venue density, UNESCO heritage access, culinary diversity, nightlife, arts funding, creative employment share, and international event density.

R1 — Cultural Venues

Unit: per 100,000 population · Direction: ↑ Higher is better · Source: OpenStreetMap; Google Places API

Raw value	Score
≤ 2	0
8	25
20	50
40	75
≥ 70	100

R2 — UNESCO World Heritage Sites

Unit: sites within 50km radius · Direction: ↑ Higher is better · Source: UNESCO World Heritage List + GIS distance analysis

Raw value	Score
0	0
1	25
3	50
6	75
≥ 10	100

R3 — Culinary Diversity

Unit: cuisine types per 100,000 population · Direction: ↑ Higher is better · Source: Google Places; Yelp; Foursquare

Raw value	Score
≤ 1	0
3	25
8	50
15	75
≥ 25	100

R4 — Nightlife Density

Unit: bars and clubs per 100,000 population · Direction: ↑ Higher is better · Source: OpenStreetMap; Google Places

Raw value	Score
≤ 3	0
10	25
25	50
50	75
≥ 80	100

R5 — Arts Funding

Unit: USD PPP per capita public arts expenditure · Direction: ↑ Higher is better · Source: Eurostat Culture Statistics; national ministries of culture

Raw value	Score
≤ 5	0
30	25
80	50
150	75
≥ 300	100

R6 — Creative Employment

Unit: % of workforce in creative industries · Direction: ↑ Higher is better · Source: UNCTAD Creative Economy Report; national labour force surveys

Raw value	Score
≤ 1%	0
3%	25
6%	50
10%	75
≥ 15%	100

R7 — International Events

Unit: per million population per year · Direction: ↑ Higher is better · Source: ICCA International Congress and Convention Association; Eventbrite

Raw value	Score
≤ 2	0
10	25
25	50
50	75
≥ 100	100

Data Sources

SLIC uses a five-tier source hierarchy. Tier 1 (city-official) is preferred; the fallback order is Tier 2 (subnational) → Tier 3 (national/international) → Tier 4 (audited secondary) → Tier 5 (analyst assessment). The source tier for each published metric is visible on the city scorecard page.

Tier	Level	Examples
1	City / metro official	Municipal open data portals, utility feeds, transit GTFS
2	Subnational official	State, provincial, or regional government data
3	National / international official	World Bank, WHO, ILO, UNESCO, OECD, WIPO, UN Agencies
4	Audited secondary & experimental	OpenAQ, Copernicus CAMS, M-Lab NDT, satellite remote sensing
5	Analyst assessment	SLIC analyst cross-reference and lived-experience review

Primary Source Organisations

- World Bank — WDI, B-READY, PovcalNet, PPP conversion factors
- World Health Organization — GHO OData API, HAQ Index, JMP water/sanitation
- International Labour Organization — ILOSTAT (employment, hours, wages)
- OECD — PISA, Health Statistics, Affordable Housing, Income Distribution
- UN Population Division — birth rates, demographic projections
- UNESCO Institute for Statistics — education enrollment, cultural indicators
- UNODC — International Homicide Statistics
- UNDP — Human Development Reports, Gender Inequality Index
- IEP — Global Peace Index
- Freedom House — Freedom in the World annual scores
- V-Dem Institute — Liberal Democracy Index
- Pew Research Center — Global Restrictions on Religion Index
- WIPO — Patent Application Statistics
- IMF — World Economic Outlook GDP data
- IRENA — Renewable capacity and generation statistics
- IQAir / WHO — PM2.5 ambient air quality annual averages
- ILGA World + Equaldex — LGBTQ+ legal environment composite
- World Values Survey + Gallup — social trust indicators
- ICCA — International Congress and Convention Association statistics
- Numbeo — Cost of living, housing, and consumer price surveys
- Crunchbase / PitchBook / Dealroom — startup and VC investment data
- Ookla + M-Lab — broadband speed measurements
- OpenStreetMap / CycloSM — cycling and pedestrian infrastructure
- Walk Score — walkability index

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- OpenAQ / Copernicus CAMS — supplementary air quality monitoring

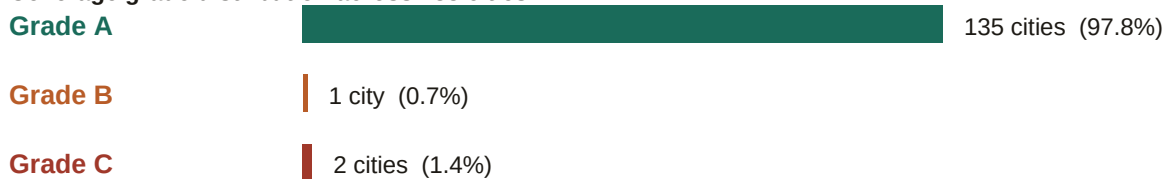
8.

Coverage Grades

SLIC does not impute missing values. When indicators are unavailable, they are excluded from the pillar aggregation and a coverage penalty is applied. The coverage grade is published alongside every city score.

Grade	Condition	Penalty	Interpretation
A	≥ 6 of 7 indicators present (or ≥ 5 of 5–6 indicator pillar)	None	Full confidence in pillar score
B	≥ 4 indicators present	–5 points	Score is reliable but interpret with moderate caution
C	≤ 3 indicators present	–15 points, flagged provisional	Treat as indicative; significant data gaps exist

Coverage grade distribution across 138 cities



The penalty is applied after the AMPI computation at the pillar level and is floored at zero. Overall coverage grade reflects the worst pillar grade for that city. Cities are never hidden because of low coverage — low coverage is surfaced, not suppressed.

9.

Published City Rankings

All 138 published cities, sorted by overall SLIC score. Scores are rounded to one decimal place. Ties are allowed — cities with the same rounded score share the same rank number. Pillar column headers: G = Growth, V = Viability, Cap = Capability, Com = Community, Cr = Creative.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
1	Raleigh	United States	66.6	70.1	84.1	79.1	43.5	62.0	A
2	Kaohsiung	Taiwan	66.1	63.2	86.5	90.6	48.1	56.9	A
3	Montreal	Canada	65.7	57.3	93.3	75.3	64.4	54.1	A
3	Singapore	Singapore	65.7	50.4	87.4	94.1	47.8	73.3	A
5	Minneapolis	United States	65.6	66.1	84.1	79.1	43.5	62.0	A
6	Pittsburgh	United States	65.3	71.5	84.1	79.1	43.5	55.7	A
6	Abu Dhabi	United Arab Emirates	65.3	65.4	92.6	62.0	78.8	46.2	A
8	Dubai	United Arab Emirates	65.0	64.4	92.6	62.0	78.8	46.2	A
9	Adelaide	Australia	64.1	60.3	89.5	87.1	53.2	49.2	A
10	Melbourne	Australia	63.3	49.1	89.5	87.1	53.2	60.5	A
11	Manama	Bahrain	63.0	82.6	79.7	57.8	62.9	42.5	A
12	Ottawa	Canada	62.7	49.3	93.3	75.3	64.4	54.1	A
12	Hobart	Australia	62.7	60.3	89.5	87.1	53.2	45.1	A
14	Graz	Austria	62.1	57.9	87.1	74.4	70.7	41.4	A
14	Perth	Australia	62.1	52.9	89.5	87.1	53.2	51.2	A
16	Taipei	Taiwan	62.0	43.7	89.5	89.7	48.1	69.4	A
17	Eindhoven	Netherlands	61.8	59.6	97.6	71.8	60.9	42.8	A
18	Brisbane	Australia	61.6	49.1	89.5	87.1	53.2	54.4	A
19	Haifa	Israel	60.6	64.1	86.1	64.7	41.6	54.7	A
20	Chicago	United States	59.8	48.4	84.1	79.1	43.5	62.0	A
21	Dunedin	New Zealand	59.0	52.6	92.6	85.7	58.4	39.1	A
22	Milan	Italy	58.8	88.5	85.8	66.9	50.8	29.4	A
23	Lyon	France	58.5	47.5	97.9	63.9	72.0	45.1	A
24	Valparaiso	Chile	58.3	56.3	72.7	83.8	49.8	44.6	A
25	Christchurch	New Zealand	58.2	46.3	92.6	85.7	58.4	43.6	A
26	Khobar	Saudi Arabia	57.4	88.5	96.3	78.9	42.8	24.1	A
27	Jeddah	Saudi Arabia	57.2	87.3	96.3	78.9	42.8	24.1	A
28	Braga	Portugal	57.0	54.5	95.7	68.8	47.3	43.8	A
28	Riyadh	Saudi Arabia	57.0	86.1	96.3	78.9	42.8	24.1	A
30	Jeju City	South Korea	56.3	53.8	99.2	85.8	30.0	50.6	A
31	New York	United States	56.2	23.9	85.4	82.0	71.5	75.3	A
32	London	United Kingdom	56.1	30.1	82.8	78.4	74.2	59.3	A
33	Copenhagen	Denmark	55.9	26.0	97.9	71.3	89.3	61.5	A
34	Sydney	Australia	55.6	31.0	89.5	87.1	53.2	63.9	A
34	Wellington	New Zealand	55.6	36.8	92.6	85.7	58.4	48.8	A
36	Torun	Poland	55.5	66.7	79.5	65.1	51.2	31.9	A
37	Vienna	Austria	55.3	40.2	87.1	74.4	70.7	41.4	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
38	Santiago	Chile	55.2	47.2	72.7	83.8	49.8	44.6	A
38	Zurich	Switzerland	55.2	33.7	97.6	72.7	72.8	49.4	A
40	Busan	South Korea	55.1	50.8	99.2	85.8	30.0	50.6	A
41	Suwon	South Korea	55.0	50.5	93.1	85.8	30.3	50.6	A
41	Incheon	South Korea	55.0	50.5	99.2	85.8	30.0	50.6	A
41	Toronto	Canada	55.0	32.4	93.3	75.3	64.4	54.1	A
44	Helsinki	Finland	54.9	33.8	91.5	81.0	58.2	52.9	A
45	Katowice	Poland	54.8	63.9	79.5	65.1	51.2	31.9	A
46	Porto	Portugal	54.5	47.5	95.7	68.8	47.3	43.8	A
47	Tallinn	Estonia	53.6	45.5	90.1	58.4	38.6	57.8	A
47	Auckland	New Zealand	53.6	30.6	92.6	85.7	58.4	52.9	A
49	Gdansk	Poland	53.4	58.2	79.5	65.1	51.2	31.9	A
50	Vancouver	Canada	53.3	29.3	93.3	75.3	64.4	54.1	A
50	Tokyo	Japan	53.3	39.7	94.2	76.8	51.6	44.2	A
52	Tel Aviv	Israel	53.1	41.4	86.1	64.7	41.6	54.7	A
53	Fukuoka	Japan	50.9	57.0	94.5	67.4	25.8	42.3	A
53	Bratislava	Slovakia	50.9	54.1	89.7	49.7	51.5	35.3	A
53	Krakow	Poland	50.9	49.6	79.5	65.1	51.2	31.9	A
56	Bologna	Italy	50.8	50.9	88.2	66.9	50.8	29.4	A
57	Amsterdam	Netherlands	49.9	31.4	97.6	71.8	60.9	42.8	A
58	Hiroshima	Japan	49.1	57.0	94.5	67.4	25.8	37.3	A
58	Sapporo	Japan	49.1	57.0	94.5	67.4	25.8	37.3	A
60	Doha	Qatar	49.0	53.9	73.9	54.4	36.5	37.6	A
61	Kuwait City	Kuwait	48.8	87.2	69.2	65.0	40.5	19.9	A
62	Kobe	Japan	48.7	55.6	94.5	67.4	25.8	37.3	A
63	Bangkok	Thailand	47.4	43.4	76.6	56.4	55.9	30.0	A
64	Port Louis	Mauritius	47.2	54.6	76.2	47.0	26.2	45.4	A
64	Melaka	Malaysia	47.2	63.0	75.3	52.4	31.4	31.5	A
66	Kuching	Malaysia	47.0	61.3	73.7	52.4	32.2	31.5	A
67	Kota Kinabalu	Malaysia	46.9	61.3	77.9	52.4	31.1	31.5	A
68	Salalah	Oman	46.5	72.0	71.1	42.0	35.9	30.6	A
69	Paris	France	46.3	23.1	97.9	63.9	72.0	45.1	A
69	Montevideo	Uruguay	46.3	49.6	59.5	74.6	50.5	24.4	A
71	George Town	Malaysia	46.1	57.2	73.7	52.4	32.2	31.5	A
72	Tianjin	China	45.6	40.6	98.6	71.9	32.4	37.1	A
73	Muscat	Oman	45.5	63.6	71.1	42.0	35.9	30.6	A
74	Bucharest	Romania	44.9	53.6	81.4	50.6	27.6	34.8	A
75	Budapest	Hungary	44.7	42.9	93.1	55.5	52.3	26.2	A
76	San Juan	Puerto Rico	44.6	35.3	59.2	68.3	61.3	45.8	B
76	Cordoba	Argentina	44.6	65.5	69.2	78.6	49.0	10.2	A
78	San Jose	Costa Rica	44.4	40.2	43.5	66.0	54.6	35.9	A
79	Chongqing	China	44.2	36.5	75.2	71.9	33.0	37.1	A
80	Kuala Lumpur	Malaysia	44.1	48.9	73.7	52.4	32.2	31.5	A
81	Belgrade	Serbia	44.0	43.4	74.8	53.2	39.7	30.3	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
82	Chengdu	China	43.8	35.4	77.0	71.9	33.2	37.1	A
83	Hat Yai	Thailand	43.5	55.1	66.7	56.4	36.4	22.8	A
84	Male	Maldives	42.4	54.1	72.1	39.8	26.5	35.4	A
85	Guangzhou	China	41.9	33.0	98.6	71.9	32.4	37.1	A
86	Phuket	Thailand	41.5	46.6	68.0	56.4	36.4	22.8	A
87	Panama City	Panama	41.4	55.5	44.0	64.2	33.0	26.9	A
88	Chiang Mai	Thailand	41.1	54.3	49.7	56.4	36.4	22.8	A
89	Hangzhou	China	40.9	29.2	84.0	71.9	33.3	37.1	A
89	Buenos Aires	Argentina	40.9	50.7	69.2	78.6	49.0	10.2	A
91	Shenzhen	China	39.9	29.2	98.6	71.9	32.4	37.1	A
92	Curitiba	Brazil	38.2	49.4	45.1	69.5	23.2	27.6	A
93	Florianopolis	Brazil	37.8	47.6	45.1	69.5	23.2	27.6	A
94	Arequipa	Peru	37.7	55.9	34.1	42.3	44.0	24.7	A
95	Nizhny Novgorod	Russia	37.6	70.2	62.4	59.2	21.1	15.5	A
96	Shanghai	China	37.2	24.4	98.6	71.9	32.4	37.1	A
97	Lima	Peru	36.8	47.8	34.1	42.3	44.0	24.7	A
98	Sao Paulo	Brazil	35.0	37.6	45.1	69.5	23.2	27.6	A
99	Medellin	Colombia	34.4	52.4	28.0	66.3	40.0	19.6	A
100	Kigali	Rwanda	33.7	54.3	47.2	15.8	53.3	24.8	A
101	Rabat	Morocco	33.4	45.5	59.9	34.1	28.9	18.8	A
101	Windhoek	Namibia	33.4	41.6	30.5	32.1	46.9	25.4	A
103	Casablanca	Morocco	33.2	44.3	59.9	34.1	28.9	18.8	A
103	Cebu City	Philippines	33.2	55.0	38.2	40.4	30.9	17.5	A
105	Bogota	Colombia	33.0	45.1	28.0	66.3	40.0	19.6	A
106	Makati	Philippines	32.8	50.1	38.2	40.4	30.9	17.5	A
107	Surabaya	Indonesia	32.1	63.2	40.8	35.1	27.6	17.8	A
107	Jakarta	Indonesia	32.1	56.6	40.8	35.1	27.6	17.8	A
107	Gaborone	Botswana	32.1	50.3	31.9	22.2	25.1	39.0	A
110	Moscow	Russia	30.4	35.2	62.4	59.2	21.1	15.5	A
111	Suva	Fiji	30.2	41.6	36.1	47.7	13.6	25.1	A
112	Aqaba	Jordan	28.2	41.5	66.0	46.5	10.8	17.0	A
113	Hyderabad	India	28.1	43.9	53.1	30.4	20.8	14.6	A
114	Pune	India	28.0	43.6	53.1	30.4	20.8	14.6	A
115	Bengaluru	India	27.9	42.7	53.1	30.4	20.8	14.6	A
116	Chandigarh	India	27.8	43.4	44.1	30.4	21.5	14.6	A
117	Amman	Jordan	27.0	37.5	66.0	46.5	10.8	17.0	A
118	Merida	Mexico	26.1	49.8	13.3	55.5	55.5	14.1	A
119	Kandy	Sri Lanka	25.4	51.5	40.7	33.7	15.8	11.7	A
119	Cape Town	South Africa	25.4	31.5	15.1	26.3	53.6	29.8	A
119	Colombo	Sri Lanka	25.4	48.3	40.7	33.7	15.8	11.7	A
119	Johannesburg	South Africa	25.4	31.7	15.1	26.3	53.6	29.8	A
123	Guadalajara	Mexico	25.3	45.0	13.3	55.5	55.5	14.1	A
124	Nairobi	Kenya	24.0	45.6	45.1	12.5	29.3	14.5	A
124	Thimphu	Bhutan	24.0	46.7	47.0	25.2	18.9	10.2	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
126	Accra	Ghana	23.4	53.3	35.9	14.9	21.7	17.8	A
127	Kumasi	Ghana	23.0	58.7	35.9	14.9	21.7	17.8	A
128	Chattogram	Bangladesh	22.7	51.3	48.3	10.8	23.3	15.7	A
128	Dhaka	Bangladesh	22.7	48.8	48.3	10.8	23.3	15.7	A
130	Pokhara	Nepal	22.1	44.3	37.6	12.2	20.9	16.6	A
130	Kathmandu	Nepal	22.1	41.7	37.6	12.2	20.9	16.6	A
132	Mexico City	Mexico	22.0	32.1	13.3	55.5	55.5	14.1	A
133	Da Nang	Vietnam	20.2	51.0	—	44.3	—	20.8	C
134	Phnom Penh	Cambodia	15.4	41.9	—	18.2	—	36.3	C
135	Dakar	Senegal	12.4	38.8	0.7	10.6	45.1	26.7	A
136	Islamabad	Pakistan	11.6	47.2	32.7	8.1	15.3	4.4	A
137	Karachi	Pakistan	11.5	48.0	32.7	8.1	15.3	4.4	A
138	Lahore	Pakistan	11.4	48.3	32.7	8.1	15.3	4.4	A

Design Principles

Absolute scoring

Anchors are fixed at real-world thresholds. A city's score does not change because a new city joined the index. Adding city #501 cannot change cities #1–500.

Transparent and traceable

Every published score traces back to a raw data point, a normalization function, and a named source. The scoring workbook is publicly downloadable.

Penalises imbalance

The AMPI aggregation formula means a city cannot compensate for catastrophic performance in one pillar by excelling in another. Extreme imbalance is structurally penalised.

Honest about data gaps

Missing data is never imputed. Gaps are flagged with visible coverage grades and penalties. A low-coverage city is shown, not hidden.

Scalable without revision

New cities can be added to the index without recalculating existing scores. The methodology does not require retrospective revision when the city universe grows.

Anti-pattern resistant

The index is designed so that cities cannot easily game their scores by optimising for visible proxy metrics. The DI_PPP formula, AMPI penalty, and imbalance penalties are all difficult to manipulate without genuinely improving urban conditions.

Notation Glossary

Symbol / Term	Definition
c	City or functional urban area being evaluated
m	Metric index (individual indicator within a pillar)
p	Pillar index (one of the five thematic pillars)
$x_m(c)$	Raw observed metric value for city c on metric m
$s_m(c)$	Normalized metric score for city c on metric m, range [0, 100]
μ	Arithmetic mean of normalized scores within a pillar or across pillars
σ	Population standard deviation of scores within a pillar or across pillars
cv	Coefficient of variation: σ / μ
AMPI	Adjusted Mazziotta–Pareto Index: $\mu - \sigma^2/\mu$
p05_m, p95_m	5th and 95th percentile bounds for metric m, frozen at first publication
DI_PPP(c)	PPP-adjusted monthly disposable income after essential costs for city c
$\alpha(p, m)$	Metric weight of indicator m inside pillar p
γ_m	Coverage weight for metric m (used in coverage ratio computation)
w_p	Public pillar weight (Growth 0.25, Viability 0.22, Capability 0.18, Community 0.15, Creative 0.20)
HAQ	Healthcare Access and Quality Index (IHME/Lancet)
GPI	Global Peace Index (Institute for Economics and Peace)
GII	Gender Inequality Index (UNDP)
GRI	Government Restrictions Index (Pew Research Center)
EGDI	E-Government Development Index (United Nations)
PPP	Purchasing Power Parity — conversion factor equalising purchasing power across currencies

SLIC Index V3 · slic.index · April 2026 · Published by SLIC in partnership with DEPA Thailand, PMU-A, and ReTL. Methodology questions: contact via depa.or.th or the SLIC website.